

SocialScribe: A Modular System for Audience-Targeted Social Media Generation from Long-Form Content

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Abstract—Social media platforms have emerged as critical environments for information exchange, requiring mechanisms to convert intricate academic and scientific texts into succinct messages tailored for specific audiences. This study introduces SocialScribe, an intelligent, automated system leveraging large language models to generate contextually suitable social media content from extended source material. The paper details the system’s architecture and implementation, with emphasis on adaptive prompt design, audience modeling, and tone regulation. Security and privacy issues are systematically examined, alongside strategies for minimizing unintended content generation and ensuring alignment with user requirements. A mixed-method evaluation, including performance metrics and user feedback, demonstrates SocialScribe’s effectiveness in facilitating broader outreach for researchers and organizations while maintaining message accuracy. The findings are discussed in relation to current advancements in generative artificial intelligence, and future avenues are identified for enhancing personalization, interpretability, and modality integration.

Index Terms—Generative AI, large language models (LLMs), audience modeling, prompt engineering, tone control, hallucination mitigation, data privacy, security, message fidelity, human-in-the-loop.

I. INTRODUCTION

Social platforms have become a dominant layer in how scientific knowledge is encountered, debated, and repurposed by both specialist and public audiences, shifting the locus of attention away from traditional long-form venues to streams of short-form, highly contextual content. For researchers and institutions, this shift creates a communication bottleneck: distilling complex, technical materials into concise, audience appropriate posts that preserve fidelity, signal uncertainty, and comply with platform constraints while competing for attention in dynamic feeds. At the same time, the rise of generative models offers an opportunity to improve reach and consistency, provided that systems are designed for audience adaptation, safety, and provenance rather than generic summarization alone [1]–[3]. Large language models (LLMs) have rapidly advanced abstractive summarization, controllable writing, and style transfer, enabling transformations of long-form sources into targeted messages. Recent pipelines incorporate multi-source inputs, salience control, and personalization to reflect the needs and prior knowledge of

distinct audience segments, moving beyond one-size-fits-all summaries toward audience-aware reframing. However, known risks—hallucinations, overgeneralization of scientific claims, and uneven adherence to source evidence—require multi-layer controls, transparent attribution, and human oversight if these systems are to be adopted for public-facing research communication. Bridging this gap, SocialScribe proposes a modular pipeline that couples multimodal analysis, explicit preference capture, hybrid LLM generation, and human-in-the-loop verification to convert scholarly inputs into platform-ready posts tuned to voice, audience, and platform norms [4]–[7].

This paper makes three contributions. First, it presents a system architecture that cleanly separates ingestion/analysis, preference contextualization, and generation, enabling targeted scaling and ablation. Second, it introduces an audience adaptation workflow that translates explicit user preferences (tone, register, audience, platform) into prompt fields and constraints, while using a content graph to preserve factual alignment and provide hooks for attribution. Third, it outlines an evaluation approach that mixes automatic metrics for content fidelity and constraint satisfaction with human interaction evaluations that capture trust, usability, and willingness to share—constructs of particular importance for public science communication. Taken together, these contributions are a step toward reliable, audience-sensitive dissemination of complex findings on social platforms [8], [9].

II. RELATED WORK

A. Summarization and Audience Adaptation:

Early approaches to summarization emphasized extractive methods and template-based generation; transformer-based LLMs now support abstractive, controllable summarization at scale. Multi-source and meeting summarization pipelines have begun to incorporate upstream persona extraction or audience modeling to adjust salience and level-of-detail, demonstrating measurable gains in informativeness and relevance relative to context-agnostic baselines. Controllable summarization for distinct expert/non-expert audiences is often achieved with prompt conditioning and readability targets, with evidence that preference-based evaluation correlates better with human

judgments than n-gram metrics, reinforcing the need for human-centred objectives in scientific reframing tasks [4]–[6], [10].

B. Science Communication on Social Platforms:

Social media is increasingly used by scientists, institutions, and funders to disseminate findings, engage communities, and support transparency, though much activity remains oneway and underutilizes opportunities for dialogue and audience participation. Studies highlight both the potential for visibility and the need for formats that balance accuracy, accessibility, and engagement, particularly in post-pandemic digital contexts and within national research ecosystems. This motivates systems that can tailor tone, vocabulary, and calls-to-action to platform norms while preserving evidential integrity [1], [2], [11].

C. Methodological Trends in Summarization:

Recent surveys map the evolution from statistical and neural extractive methods to LLM-based abstractive pipelines that integrate retrieval augmentation, hierarchical planning, and human feedback. Key trends include (i) multi-source fusion with redundancy and contradiction handling, (ii) hierarchical planning for long inputs, and (iii) evaluation beyond ROUGE to human preference modeling and interaction outcomes. Clinical and timeline summarization work illustrates domain-specific prompting and hybrid architectures (e.g., VAE+LLM) to capture temporal coherence and point-of-view, relevant to transforming research narratives into lay-accessible threads while preserving voice and context [6], [10], [12], [13].

D. Risks: Overgeneralization and Hallucinations:

LLMs may overgeneralize scientific conclusions or produce unsupported statements, especially under domain shift or incomplete retrieval. Recent analyses document generalization bias and propose mitigation via grounding, entailment checks, and uncertainty-aware prompting, underscoring that safety and fidelity are preconditions for responsible public dissemination. Audience-aware reframing also risks persuasive misframing if style is decoupled from evidence; thus, attribution and human verification are essential design elements [6], [7].

E. Positioning SocialScribe:

Compared to generic summarization tools, SocialScribe foregrounds (i) explicit audience modeling and platformware constraints, (ii) structured content graphs for grounding and attribution, and (iii) a hybrid generation controller with uncertainty gating and mandatory human verification in sensitive contexts. By integrating these facets, the system targets the operational realities of science communication—voice preservation, provenance, safety, and platform fit—rather than optimizing solely for brevity or coherence [1], [4], [6].

III. CHALLENGES

We formalize six challenge dimensions relevant to automated post generation: (C1) Subjectivity and Fidelity: differing notions of “good” posts; need for traceable alignment with source claims; summary faithfulness and coverage audits. (C2) Style and Audience Adaptation: tone, register, rhetorical devices, and readability level controls with platform norms. (C3) Privacy and IP: handling unpublished and sensitive materials; data minimization, redaction, and consent. (C4) Factuality and Safety: hallucinations, misattribution, and toxicity; need for detection, retrieval grounding, and review. (C5) Platform Fit: constraints on tokens, links, media, and interactive features; automatic layout and threading. (C6) Trust and Ownership: preserving authorial voice, clear provenance, and IP boundaries to support adoption [4], [5], [7], [9].

IV. METHODOLOGY AND ARCHITECTURE

A. Pipeline Overview:

SocialScribe comprises three phases: (1) Multimodal Source Analysis; (2) User Preference & Contextualization; (3) LLMPowered Generation with Human Verification. The modular design enables independent scaling and targeted ablations.

B. Multimodal Source Analysis:

We preprocess text (PDF to UTF-8; subtitle parsing; sentence consolidation) and apply NLP: tokenization, part-of-speech tagging, dependency parsing, and topic modeling (e.g., LDA) to extract domain terms and themes. Narrative understanding identifies entities, character networks via social network analysis, and key events. Outputs form a structured content graph: entities, claims, evidence spans, salience scores, and candidate hooks.

C. Preference & Contextualization:

An interface captures tone (formal, friendly, humorous), audience segment (researchers, practitioners, public), platform target, and priority themes. A preference model translates selections into prompt fields: style directives, reading level, CTA policy, platform constraints (character limits, threading), hashtag/link policies, and safety constraints. Prompts include content graph snippets and inline attribution cues to nudge grounded generation.

D. LLM Generation and Hybrid Control:

We adopt a hybrid controller: a lightweight LLM handles compression, outline synthesis, and constraint satisfaction; a larger LLM performs final drafting with safety-steered decoding. Optional RAG provides citations and snippets for grounding. An uncertainty gate (entropy/top-k variance) flags risky spans for human review or triggers regeneration with

stricter constraints. Human verification is mandatory for sensitive deployments [9].

TABLE I: RELATED WORK ANALYSIS AND RESEARCH GAPS (MODEL COLUMN REMOVED FOR BETTER READABILITY).

The surveyed literature shows a growing interest in systems requirements, and broad narrative domains. Studies on social

SNO	Author & Citation	Experiment	Outcome	Research Gap
1	Eldamnhory, M. (2024). <i>Academify-Lite</i> et al.	Developed Academify-Lite, GPT-4 powered tool; conducted user study.	Reduced researchers' time in post creation; improved clarity & engagement.	Limited to X (Twitter) sharing; needs broader platform integration.
2	Zhou, P., Xu, T., Yin, Z., Liu, D., Chen, E., Lv, G., & Li, C. (2020). et al.	Proposed framework combining visual detection + textual cues; experiments on real-world datasets.	Outperformed state-of-art baselines in person search & summarization.	Difficulty in varying background/pose; real-time efficiency still a challenge.
3	Zlatintsi, A., Koutras, P., Evangelopoulos, G., et al. (2017). <i>COGNIMUSE</i> . EURASIP JIVP. et al.	Built COGNIMUSE multimodal dataset; tested saliency-based summarization models	Validated usefulness of COGNIMUSE for summarization: summaries rated high.	Other datasets lack cross-modal long annotations, expansion needed
4	Shen, S., & Garg, A. (2025). <i>Adapting LLMs for Movie Domain with Narrative Understanding Tasks</i> . et al.	Constructed narrative understanding dataset; tested adaptation strategies on LLMs	Improved narrative comprehension by LLMs; highlighted trade-offs in adaptation	Generalization vs. specialization trade-off unresolved.
5	Do, T.T.H., Tran, Q.H.B., & Tran, Q.D. et al.	Applied character network analysis (social strength of characters) to summarize movies.	Summaries preserved storyline coherence; effective indexing in movie datasets.	Accuracy and processing time of social-network methods unsatisfactory
6	Evangelopoulos, G., Zlatintsi, A., Potamianos, A., et al. (2013). et al.	Designed multimodal saliency model (audio, visual, textual); fusion tested on movie DB	Produced aesthetically high-quality summaries; strong multimodal fusion results.	Fusion models need semantic & affective context integration.
7	Darling et al. (2013). <i>Use of Twitter in Scientific Communication</i> . et al.	Analyzed Twitter's role in research networking & pre-publication idea exchange.	Twitter useful but risks: oversimplification & ownership disputes.	Risks of misinformation and exclusion not fully solved.
8	Peoples et al. (2016). <i>Twitter Activity & Citation Correlation</i> . et al.	Studied Twitter metrics vs. article citation counts in ecological journals.	Strong correlation between tweets and citation counts.	Causation between engagement and impact unclear.
9	Weissburg et al. (2017). <i>Role of Influencers in ML Visibility</i> . et al.	Studied ML paper visibility via Twitter influencers; citation impact observed.	Influencers significantly boost ML research reach and citations.	Influence dynamics lack deeper modeling of credibility.
10	Chatman, S. (1980). <i>Narrative Theory</i> (as cited in Shen & Garg 2025). et al.	Applied narrative theory to design LLM tasks (story vs. discourse)	Framework for narrative-based adaptation of LLMs established	Scalability and generalizability to broader narratives needed

that simplify complex content and tailor it for specific audiences, particularly through multimodal summarization and narrative-aware language models. While these methods have improved clarity, engagement, and content fidelity, they remain limited in handling diverse platforms, real-time

media use in research communication also reveal increased visibility but raise concerns about misinformation, credibility, and weak causal links between online engagement and academic impact. Collectively, prior work highlights the need

for scalable, context-sensitive, and trustworthy systems for academic content dissemination.

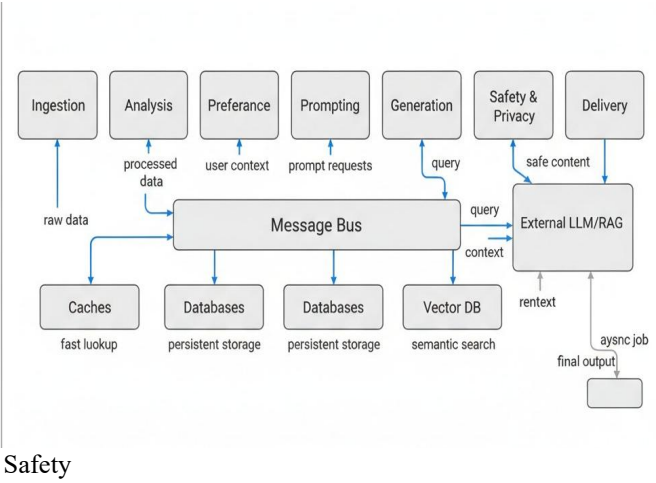
V. IMPLEMENTATION

A. Data and Libraries:

PDF parsing (GROBID/PyMuPDF), SRT/VTT parsing, spaCy or Stanza for POS/NER/dependencies, genism for LDA, NetworkX for SNA, and regex-based cleaners. For generation: OpenAI/Anthropic/Local LLM via API with retry/backoff. Safety: PII detectors (NER with custom patterns), toxicity classifiers, prompt policies.

B. Prompt Engineering Templates:

We use slot-based templates parameterized by audience, tone, and platform. Content graph snippets provide salient claims and source spans. Soft constraints enforce reading level and CTA placement; hard constraints enforce character limits.



Safety

Fig. 1. System architecture with modular services and hybrid LLM controller instructions prohibit unverifiable facts and require hedging or omission for uncertain content.

C. Hybrid Controller and Caching:

A controller routes tasks: compress→outline→draft→refine. Token budgets are enforced per stage. Caching stores embeddings, LDA topics, and outlines keyed by document hash to reduce recompilation. Batch inference amortizes API overhead; streaming decoding reduces perceived latency.

D. Performance Engineering:

We track p50/p90/p99 latency, tokens/request, cache hit rate, and throughput. Parallel content graph extraction and asynchronous generation improve utilization. Table II shows a reference budget.

E. Evaluation Plan:

Automatic: ROUGE/BERT Score for content coverage; style metrics (readability, sentiment, formality); constraint satisfaction rate; platform-fit (length, link/hashtag validity).

Safety: toxicity, PII leakage, hallucination-flag rate (uncertainty proxy). Human: HIE protocol measuring fidelity, usefulness, tone fit, trust, and willingness to share in realistic tasks [5], [7], [9].

TABLE II: REFERENCE LATENCY BUDGET AND COSTS (ILLUSTRATIVE).

Stage	Latency (Ms)	Tokens
Parsing & NLP	300–800	–
Topic & SNA	200–600	–
Compress & Outline (small LLM)	400–900	0.8k
Draft (large LLM)	1200–2500	1.2k
Safety & Postproc	150–400	–
End-to-end (p50)	2300–4500	2.0k

VI. SECURITY AND PERFORMANCE

A. Privacy and IP Protections

SocialScribe enforces data minimization and scoped processing by default, retaining only what is necessary for analysis and generation and discarding transient artifacts as soon as downstream tasks complete to reduce exposure surface area. Inputs can be processed in a split mode in which sensitive source documents remain on the client or a controlled enclave, and only derived features (e.g., anonymized entity graphs, topic vectors) are transmitted to the service, thereby limiting identifiable content flow across boundaries.

Personally identifiable information (PII) is mitigated via layered detection: pattern-based rules, domain-tuned named entity recognition, and context classifiers flag direct and quasi-identifiers; flagged spans are masked or pseudonymized before storage or external model calls unless an authorized policy explicitly permits passthrough for designated projects. All traffic uses TLS 1.3 with modern ciphers, and data at rest is encrypted with customer-managed keys where available; fine-grained access control lists (ACLs), least-privilege service roles, and comprehensive audit logs establish accountability and facilitate incident response and regulatory inquiries.

For higher-assurance deployments, confidential computing (TEEs for inference and orchestration), private VPC endpoints, and bring-your-own-key (BYOK) reduce trust in cloud operators; optional differential privacy can be applied to aggregate telemetry to prevent re-identification in small cohorts. Intellectual property is preserved through explicit terms: inputs remain the contributor’s property; outputs are licensed to the contributor for reuse; prompts/outputs are excluded from model training by default unless the owner provides written, revocable consent; and retention windows and deletion SLAs are clearly surfaced in the interface and enforced in back-end policies [9], [10].

B. Safety Guardrails:

The system employs multi-layer guardrails spanning preprompt policies, retrieval-time constraints, decoding-time

controls, and post-generation filtering to reduce harmful or ungrounded outputs without over-blocking useful content. Pre-prompt policies specify tone, allowed/forbidden content categories, source-of-truth requirements, and audience appropriateness; these constraints are injected into all stages, including outline and drafting, to steer behavior early rather than relying solely on post hoc filtering. When retrieval augmentation is used, citation-bearing snippets and structured claims are passed to the generator with attribution tags; the model is instructed to limit assertions to retrieved or user-provided facts and to omit or hedge uncertain claims, lowering the risk of extrinsic hallucinations. Decodingtime controls include conservative temperature/top-p tuning for factual segments, stopword and regex guards for prohibited content, and entropy- or variance-based uncertainty gates; segments exceeding uncertainty thresholds are regenerated with stricter constraints or routed to human review.

Post-generation filters (toxicity, hate/abuse, personal data leakage, product claims) are tuned to minimize false positives for scientific context while reliably catching risky strings; flagged spans trigger automatic redaction or an interactive editor prompt with rationale and suggested rewrites, preserving authorial control. Safety evaluations combine static tests (prompt suites, red teaming checklists), scenario-based user studies, and longitudinal monitoring of user-reported issues; metrics include policy violation rate, groundedness/attribution scores, uncertainty-flag coverage, and reviewer agreement, aligning with emerging guidance for human interaction evaluations (HIEs) [13], [14].

C. Scalability and Performance Engineering:

To meet latency and throughput goals, SocialScribe separates I/O-bound analysis from token-bound generation and orchestrates both with asynchronous pipelines, batching, and caching at multiple layers. A hybrid controller routes compression and outline steps to a smaller, faster model and reserves the final drafting for a larger model only when required by quality thresholds, trimming tokens and cost without degrading output fidelity for typical posts.

Prompt, embedding, and semantic-response caching—keyed by document hash, preference vector, and platform target—can eliminate redundant calls for repeated or similar tasks; approximate semantic caching tolerates paraphrase by using vector similarity to reuse high-quality drafts with minimal edits. Batch inference amortizes overhead for short calls (e.g., compression, classification), while streaming decoding improves time-to-firsttoken for the final draft; circuit breakers and fallback paths degrade gracefully to single-model generation when upstream capacity is constrained.

Resource governance enforces per-tenant quotas and token budgets with backpressure on queues, while autoscaling policies consider concurrency, average output length, and cache hit rate to right-size replicas in real time. A performance SLO budget partitions end-to-end latency across preprocessing, analysis, generation, and safety passes; continuous profiling tracks p50/p90/p99 latency, tokens per request, batch size efficiency, and cache hit ratio to guide optimization and cost

control. Finally, observability couple’s metrics with structured traces and redaction-aware logs to facilitate root cause analysis without risking leakage of sensitive content, aligning operational excellence with privacy-by-design [10], [22].

VII. DISCUSSION AND FUTURE WORK

A. Explainability, Attribution, and Provenance

A recurring barrier to adoption in research communications is the opacity of generation: stakeholders need to understand why a particular claim or phrasing appears and how it maps back to the source material. Beyond simple extractive highlights, SocialScribe can expose layered explanations that include (i) input-span attribution for each generated sentence or clause, (ii) claim-level entailment checks against retrieved snippets, and (iii) model-uncertainty indicators to contextualize confidence in factual segments.

Emerging attribution techniques—ranging from entailmentstyle verification and attention-guided attributions to explicit URL/citation generation—can be integrated into the drafting loop to both steer generation and document provenance, reducing extrinsic hallucinations while improving user trust in the final post. Concretely, future iterations will explore attributive masking and saliency-oriented methods that provide token-totoken rationales without overwhelming authors, complemented by compact, human-readable evidence panels embedded in the editor [1]–[7].

B. Human-in-the-Loop Evaluations and Trust

Static benchmarks capture a narrow slice of safety and usefulness; social media communication is interactive, contextual, and audience dependent. Human Interaction Evaluations (HIEs) better reflect real-world risks by measuring overreliance, persuasion, and context drift under realistic tasks and time pressure. SocialScribe’s evaluation roadmap emphasizes mixed methods: blinded expert ratings for fidelity and tone fit, A/B engagement tests for platform performance, and structured HIE protocols that probe failure modes such as subtle misattribution, persuasive mis framing, or overconfident claims. Iterative HITL loops, where authors review uncertainty flagged spans and accept/rewrite with rationale capture, provide high-signal feedback for preference modeling and guardrail tuning—critical for sustaining trust as topics, platforms, and policies evolve [8]–[12].

C. Cross-Modal Grounding and Content Robustness:

Scientific communication often involves figures, tables, and videos; capturing their semantics remains challenging for text-only pipelines. Extending SocialScribe with crossmodally grounding—e.g., extracting captions, chart summaries, and salient claims from figures, or event/role structure from videos—enables posts that reference multimodal evidence while maintaining attribution.

Recent work on grounding multimodal misinformation and manipulation underlines the need for precise localization of misleading elements across modalities; adopting similar grounding mechanisms can help SocialScribe highlight evidence boundaries and avoid overgeneralization from visual

cues or partial transcripts. Future work includes a multimodal fusion layer that adaptively weights modalities by domain and platform, coupled with visual-text attribution links for transparent evidence trails [13]–[16].

D. Personalization, Preference Learning, and Style Control:

Hand-tuned preference panels are a starting point but do not scale to fine-grained voice and audience nuances. SocialScribe will explore implicit preference learning from edit histories and engagement signals, with privacy-preserving aggregation to avoid overfitting to individuals or leaking sensitive intent. Style controllers that target readability bands, rhetorical devices, and call-to-action placement can be optimized per platform while preserving message fidelity, supported by constrained decoding and post-edit alignment that preserves the author’s distinctive phrasing. Longitudinal studies should assess whether personalization increases perceived authenticity without drifting from source evidence, using attribution and uncertainty signals to mark the limits of acceptable stylistic transformation [2], [10], [11], [17], [18].

E. Governance, Policy Alignment, and Organizational Integration

As institutions formalize policies on AI-assisted communications, provenance, consent, and data handling become first-class requirements. SocialScribe’s governance track will integrate policy-aware prompts, audit-ready logs, and configurable approval workflows that enforce role-based review before publication in sensitive contexts.

Watermarking or signed provenance metadata can signal AI assistance to downstream platforms where appropriate, while clear opt-in consent and data retention controls align operations with privacy guidelines and sectoral regulations. Organizational deployments will also require domain-specific guardrails (e.g., medical or financial claims), coupled with curated retrieval sources and tiered confidence thresholds to reduce liability from over general claims [7], [17]–[20].

F. Limitations:

Despite multi-layer guardrails, residual risks remain subtle hallucinations under domain shift, biased framing when training data are unbalanced, and degraded grounding when sources are incomplete or paywalled. Human oversight is still necessary for high-stakes communications; future improvements should focus on better uncertainty calibration, richer attribution UX, and red-teaming for platform-specific exploit vectors to continually surface and remediate blind spots [7], [10], [20], [23].

VIII. CONCLUSION

Social Scribe is a modular system designed to convert complex academic content into platform-ready, audience-focused social media posts. It combines source analysis to extract key themes, a preference engine that adapts tone and format to user and platform needs, and a layered LLM pipeline that ensures accuracy, speed, and safety. The architecture separates analysis from generation for scalable

performance and uses caching, routing, and filtering for efficiency and security. Although issues like subjective quality standards, platform variation, and subtle hallucinations remain, future improvements aim to enhance attribution, multimodal grounding, and privacy-preserving personalization to support reliable, transparent science communication at scale.

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